



The Shopify + Stripe for the **Agentic AI Economy**

In 2026, the next billion ecommerce transactions will not be initiated by humans browsing catalogs. They will be executed by **autonomous AI shopping agents** (ChatGPT, Claude, Perplexity, and Enterprise Procurement Bots) buying on behalf of users and companies. **Storefront for Machines** converts any merchant into a machine-readable, AI-discoverable storefront in 60 seconds, with autonomous settlement powered by **Razorpay**.

The Trillion-Dollar Problem

Trapped Behind Human Interfaces

DOM Barrier

99% of ecommerce stores are designed for human visual perception. Dynamic JavaScript, heavy CSS, and bot-blocking CAPTCHAs prevent autonomous reasoning models from understanding products, real-time availability, and options.

Stale Prices & Price Drift

Risk to GMV

Third-party aggregators (Amazon, Flipkart, Google Shopping cache) display outdated cached prices. Autonomous buyers attempting checkout on stale data cause order cancellations, disputes, or merchant margin loss.

The 2FA / OTP Payment Wall

Checkout Drop-Off

Traditional payment gateways require SMS OTPs or physical biometric verification on personal phones that AI shopping agents do not possess. Autonomous checkout fails 100% of the time at the final payment step.

Unchecked Runaway Spending

Safety Risk

Without mathematical spending guardrails, giving autonomous agents financial autonomy risks catastrophic hallucination orders, duplicate charging over unstable networks, and regulatory non-compliance.

The Solution: Non-Technical Merchant Onboarding

Merchants are business operators, not programmers. Storefront for Machines enables complete store onboarding in 4 steps with zero technical friction:

1. Enter Website URL

Autonomous Crawler

Direct crawler scans merchant domain, extracts product names, variants, high-res images, pricing tags, and real-time inventory.

2. Connect Store

1-Click OAuth Sync

Live integration with Shopify, WooCommerce, and Magento stores synchronizing inventory counts, prices, and variant matrices.

3. Upload Catalogue PDF

Document Intelligence

Optical layout parser extracts tabular product data, SKUs, and specifications from brochures and offline wholesale catalogs.

4. Start Automated AI Setup

Instant Indexing

Generates Schema.org JSON-LD data feed, Model Context Protocol (MCP) catalog endpoints, and AP2 payment mandates.



The 5 Core AI Agents

- 1

Ingest Agent: Normalization & Machine Feed Generator

Crawls websites, Shopify APIs, and PDF documents. Normalizes messy data into canonical products with variant trees (size, color, RAM), category detection, and exposes live **Schema.org JSON-LD** and **Model Context Protocol (MCP)** endpoints for LLMs.

Production Ready
- 2

Integrity Agent: Price Drift & Truth Crawler

Monitors external aggregators (Amazon, Flipkart, Swiggy Instamart) in real-time. Calculates price drift percentages and confidence scores (95–99%). Flags discrepancies on an anomaly ledger and triggers 1-click automatic remediation.

Active Monitoring
- 3

Visibility Agent: Generative Engine Optimization (GEO)

Simulates buyer queries against AI models (ChatGPT, Claude, Perplexity). Evaluates price/category constraint matching, generates GEO scorecards (Brand Visibility, Accuracy, Citations), and auto-optimizes catalog feeds for Top 3 placement.

Rank #1 Target
- 4

Outreach Agent: Natural Language Intent to Cart Builder

NLP entity extraction engine parses conversational buyer signals (e.g. *"I need 100 office chairs under ₹50,000"*). Extracts intent, quantity, and budget, binds itemized products, computes GST, and verifies regulatory consent.

TRAI Compliant

Agent 5: Paymaster Agent Powered by Razorpay

AP2 Protocol

The settlement core of Agentic Commerce. Implements the **Agentic Payment Protocol (AP2)** directly on top of **Razorpay Test Mode APIs**, enabling safe, autonomous machine settlement without OTP friction.

AUTONOMOUS GATE

Orders ≤ ₹2,000

Captures via Razorpay instantly in <400ms

HUMAN APPROVAL GATE

Orders > ₹2,000

Interactive slide-out drawer with 1-click capture

ZERO DOUBLE CHARGE

SHA-256 Idempotency

Network retries return cached transaction record

Production Verification & Test Metrics

MODULE TESTED	VERIFICATION SCOPE	STATUS	PASS RATE
Ingest Agent	Multi-source crawling, PDF parsing, variant trees, Schema.org JSON-LD	Verified	18 / 18 (100%)
Integrity Agent	Aggregator truth crawler, price drift formula, anomaly ledger remediation	Verified	12 / 12 (100%)
Visibility Agent	Natural language query matching, budget constraint logic, GEO scorecards	Verified	13 / 13 (100%)
Outreach Agent	Buyer intent extraction (quantity, budget, product), TRAI consent check	Verified	11 / 11 (100%)
Paymaster Agent	Razorpay test orders, mandate spend limits, idempotency, human gate	Verified	13 / 13 (100%)
E2E Simulation	Demo Electronics Store: 500 products ingested, 100 autonomous orders	Verified	60 / 60 (100%)
DeepSeek AI Engine	Entity extraction & catalog normalization with intelligent local fallback	Active	Zero Downtime

Production Readiness: 98 / 100 • React 19, TailwindCSS, Express 5, Prisma/Neon

github.com/Hardik-droid/razorpay-hack